

Transmit Power Control for Downlink mmWave Communications in V2I Scenarios

Abstract—

*Index Terms—*component, formatting, style, styling, insert

I. INTRODUCTION

Millimeter-wave links offer extremely large bandwidth and can deliver multi-gigabit data rates, making them attractive for next-generation vehicular and V2X systems where the traffic demand continues to rise. At the same time, operating at mmWave introduces several fundamental challenges. These frequencies experience strong attenuation, are highly sensitive to blockage, and exhibit much higher path loss than traditional sub-6 GHz bands [1]. Since mmWave signals have limited ability to penetrate common materials [2], maintaining a reliable link requires forming narrow, directional beams with large antenna arrays [3]. By steering energy towards the intended TX–RX direction, beamforming compensates for the severe propagation loss and enables a stable, high-directionality link.

However, energy efficiency has become a key performance requirement for 5G and emerging 6G systems, drawing significant interest from both research and industry communities [4]. Traditionally, base stations (BSs) have been the major power consumers in wireless communication networks, even in the absence of data transmission [5]. The mmWave infrastructure is significantly more power-consuming than sub-6 GHz systems. The combination of wide bandwidths, dense traffic, and the need for high transmit power substantially increases the energy consumption. Although mmWave can dramatically enhance throughput and link quality for vehicular connectivity, these benefits come with increased device-level power consumption and growing system-level energy costs. Addressing the energy-efficiency problem through smarter beam management, adaptive gain control, is therefore essential for sustaining high-rate mmWave V2X communication without escalating energy usage.

II. RELATED WORKS

Early research on mmWave power control relies on deterministic optimization, where power levels, beam patterns, and beamwidth are treated as fixed decision variables. In uplink and downlink NOMA systems, analog beamforming is constrained by constant-modulus hardware, and joint power–beam optimization is formulated to maximize achievable sum rate. The problems are decomposed into smaller subproblems such as beam-gain allocation or analog beamformer search, and closed-form or semi-closed-form updates are used to reach suboptimal solutions [6], [7]. These approaches depend on strong assumptions such as dominant-path alignment, fixed

user grouping, or simplified multipath models, which limit scalability in environments where channels vary quickly. As mmWave deployments evolve, energy efficiency becomes a central design metric. Train–ground systems introduce long-range mmWave links with strong Doppler effects and strict QoS constraints. In [5], energy efficiency is defined as delivered data per unit energy, and fractional programming with Lagrangian updates is used to compute transmit-power profiles along the track. The resulting solutions remain tied to deterministic propagation assumptions, fixed train trajectories, and single-link analysis, leaving limited flexibility for dynamic settings.

Reinforcement learning emerges as a way to deal with the restricted adaptability of optimization-based designs. In [8], Q-learning is introduced to adjust downlink power levels in massive MIMO under jamming, using state–action exploration without explicit channel models. The method depends on predefined state features, offline training samples, and discrete action sets, which restrict its ability to handle continuous control variables. Subsequent studies adopt DQN frameworks to jointly determine beamforming directions, power levels, and interference-coordination actions in multi-cell environments [9]. States encode channel and interference information, and rewards track sum-rate improvements, but central control introduces signaling overhead and function approximation raises convergence and stability issues.

Beamwidth and power control are later treated jointly in [10], where deep RL maps discrete beamwidth–power pairs to expected rate gains. A DQN with CSI-derived states is used, and the reward becomes an instantaneous rate without discounting. This requires the discretization of inherently continuous parameters, which affects policy granularity and convergence speed. Another RL-based beam-training model [11] optimizes energy efficiency or spectral efficiency by attaching reward penalties to training overhead, but these formulations assume equal power distribution over paths and rely on simulation-driven training.

Parallel work continues to use optimization and game-theoretic approaches. Generalized Nash games model power control and beamforming as interdependent choices, where best-response updates converge to equilibrium under specific feasibility conditions [12]. Coverage analysis under beam-misalignment errors in [13] links fractional power control to SNR degradation, using statistical misalignment models that assume a dominant path. In [14], secure mmWave-NOMA systems use clustering and successive convex approximation to allocate power under eavesdropping constraints, with non-

convexity handled through iterative DC transformation. Further, multi-user coexistence between NR-U and WiGig is formulated as a MINLP in [15], and penalty dual decomposition with CCCP solves user grouping, beam coordination, and power allocation jointly, though the computational complexity remains high.

More recent work begins addressing mobility, multi-hop networks, and temporal variations. Location-aided beamwidth control uses genetic algorithms to maximize average rate under beamwidth and SNR constraints in [16]. The method depends on location information and assumes a single LOS path, limiting generality. Bootstrapped and Bayesian DQN techniques [17] improve exploration by combining bootstrap sampling and Thompson sampling for joint beam and power decisions. However, bootstrap diversity shrinks with accumulated data, and Bayesian components increase running time. On the other hand, in [18], double-RIS-assisted multi-user uplink systems introduce collaborative beamforming and user power control solved through alternating optimization, inheriting sensitivity to initialization and slow convergence.

The most recent RL contributions target cooperative and secure vehicular mmWave links. Dueling Double Deep Recurrent Q-Networks [19] integrate convolutional and LSTM layers to handle high-dimensional states and temporal correlation for joint beamforming, relay selection, and power allocation under secrecy constraints. The decision space remains discrete, limiting direct control of continuous transmit-power variables. In multi-flow multi-hop networks, joint beam and power control is solved using block coordinate descent with successive convex approximation for power and particle swarm optimization for 3D beam steering [20]. The approach converges to local solutions and has computational cost that limits real-time deployment in dynamic channels. Across these developments, the field moves from deterministic and decomposition-based optimization toward reinforcement learning and hybrid methods.

The evolution is motivated by the need to handle rapidly changing mmWave channels, multi-hop topologies, energy constraints, and mobility. Nevertheless, existing RL methods still depend on discretization, offline simulation, or centralized control, while optimization-based methods continue to rely on convex relaxations and structural assumptions. These gaps highlight the need for approaches that operate directly on hardware, adapt online, and manage coupled parameters without heavy computational overhead.

III. SYSTEM OVERVIEW

A. System Model

We consider a downlink mmWave communication scenario operating at 60 GHz, in which a Road Side Unit (RSU) communicates with a user equipment (UE). The RSU and the UE are equipped with N_R uniform linear array (ULA) antennas. Transmit power control is centralized at the BS while the UE operates with a fixed receive configuration.

The received signal at the BS is given by [21]:

$$y(t) = \mathbf{w}_{ue}(t)^H \mathbf{H}(t) \mathbf{w}_{rsu}(t) s(t) + n(t), \quad (1)$$

where $\mathbf{w}_{ue}(t) \in \mathbb{C}^{N_U \times 1}$ and $\mathbf{w}_{bs}(t) \in \mathbb{C}^{N_B \times 1}$ are the UE transmit precoder and BS receive combiner, respectively, $\mathbf{H}(t) \in \mathbb{C}^{N_B \times N_U}$ is the downlink channel matrix, $s(t) \in \mathbb{C}$ is the transmitted signal, and $n(t) \sim \mathcal{CN}(0, \sigma^2)$ is the complex Gaussian noise with zero mean and variance σ^2 .

B. Problem Formulation

Assumptions. We assume predefined and fixed beambooks $\mathcal{B} = \{\mathbf{b}_1, \dots, \mathbf{b}_{|\mathcal{B}|}\}$ and $\mathcal{U} = \{\mathbf{u}_1, \dots, \mathbf{u}_{|\mathcal{U}|}\}$ at the BS and UE, respectively, with overlapping beams of fixed width. Furthermore, the beambook indices are defined as $k_{bs} \in \{1, \dots, |\mathcal{B}|\}$ and $k_{ue} \in \{1, \dots, |\mathcal{U}|\}$, and the beambook beam angles are denoted as $\Theta_{k_{bs}}$ and $\Theta_{k_{ue}}$. The BS and UE operate under LoS conditions with aligned, parallel boresights. Extension to non-line-of-sight conditions is considered out of scope and constitutes our future work. The channel matrix $\mathbf{H}(t)$ is time-varying due to environmental dynamics and mobility, and the optimal transmit/receive beamforming indices k_{ue}^* , k_{bs}^* are unknown. Additionally, the packet size per beam search is fixed, the SNR constraint, $\mathcal{S}_{th}$, is application-specific, and given.

Problem. The objective of this work is to minimize the transmit power while ensuring that the received signal-to-noise ratio (SNR), denoted by $\mathcal{S}$, remains above a predefined threshold $\mathcal{S}_{th}$ required to maintain a reliable communication link. This is achieved by jointly and adaptively controlling the transceiver gain parameters and the transmit and receive beamforming directions selected from predefined beam codebooks.

Let the transmit power at the base station, P_{bs} , be expressed as a function of the transceiver gain parameters $\{g_1, g_2, \dots, g_N\}$. The achievable SNR depends on both the gain configuration and the selected beamforming indices at the base station and user equipment, denoted by k_{bs} and k_{ue} , respectively. The resulting optimization problem is formulated as:

$$\begin{aligned} \min_{\{g_1, \dots, g_N, k_{bs}^*, k_{ue}^*\}} \quad & P_{bs}(g_1, g_2, \dots, g_N) \\ \text{s.t.} \quad & \mathcal{S}(g_1, g_2, \dots, g_N, k_{bs}^*, k_{ue}^*) \geq \mathcal{S}_{th}, \\ & g_i^{\min} \leq g_i \leq g_i^{\max}, \quad \forall i \in \{1, 2, \dots, N\}, \\ & k_{bs}^* \in \{1, \dots, |\mathcal{B}|\}, \quad k_{ue}^* \in \{1, \dots, |\mathcal{U}|\} \end{aligned} \quad (2)$$

Here, $\mathcal{S}(g_1, g_2, \dots, g_N, k_{bs}^*, k_{ue}^*)$ denotes the achievable SNR as a function of both the transceiver gain configuration and the selected optimal transmit and receive beams. The sets $\mathcal{B}$ and $\mathcal{U}$ represent the finite beam codebooks available at the transmitter and receiver, respectively. The constraints $g_i^{\min}$ and $g_i^{\max}$ define the permissible operating range of each gain parameter based on hardware limitations.

C. Deep Reinforcement Learning

Reinforcement learning (RL) is mathematically formulated based on the Markov Decision Process (MDP) [22]. An MDP

is represented by a tuple $(\mathcal{S}, \mathcal{A}, \mathcal{R}, \mathcal{P}, \gamma)$ [23], where $\mathcal{S}$ denotes the state space, $\mathcal{A}$ the action space, $\mathcal{R}$ the reward function, $\mathcal{P}$ the state transition probability, and $\gamma \in [0, 1]$ the discount factor. At each time step t , the agent observes the current state s_t and selects an action a_t according to its policy π . Upon taking the action, the agent receives an immediate reward r_t and transitions to a new state s_{t+1} . The objective of the agent is to learn an optimal policy that maximizes the expected cumulative discounted reward, expressed as

$$R_t = r_{t+1} + \gamma r_{t+2} + \gamma^2 r_{t+3} + \dots = \sum_{k=0}^{\infty} \gamma^k r_{t+k+1}. \quad (3)$$

The action-value function $Q^\pi(s, a)$ represents the expected cumulative reward obtained by taking action a_t in state s_t and subsequently following policy π . It is defined as

$$Q^\pi(s, a) = \mathbb{E}[R_t \mid s_t = s, a_t = a, \pi]. \quad (4)$$

The goal of solving an MDP is to find the optimal policy π that maximizes the action-value function, given by

$$\pi^* = \arg \max_a Q^{\pi^*}(s, a), \quad s \in \mathcal{S}, ; a \in \mathcal{A}. \quad (5)$$

Accordingly, the optimal action-value function is expressed as

$$Q^{\pi^*}(s, a) = \max_{\pi} Q^\pi(s, a), \quad s \in \mathcal{S}, ; a \in \mathcal{A}. \quad (6)$$

In the Deep Q-Network (DQN) algorithm [24], the optimal action-value function is approximated by a neural network that maps each state s to a vector of Q-values $Q(s, \cdot; \theta)$. The target network, with parameters θ^- , is updated periodically by copying the online network parameters, yielding more stable targets. The DQN update uses

$$Y_t^{\text{DQN}} = R_{t+1} + \gamma \max_a Q(s_{t+1}, a; \theta^-). \quad (7)$$

Standard DQN uses the same values to select and evaluate actions, which leads to overestimation. Double Deep Q-Network (DDQN) [25] mitigates this by decoupling selection and evaluation. The online network selects the action, and the target network evaluates it, producing the target

$$Y_t^{\text{DDQN}} = R_{t+1} + \gamma Q(s_{t+1}, \arg \max_a Q(s_{t+1}, a; \theta_t); \theta_t'). \quad (8)$$

Here, θ and θ' denote the parameters of the online and target networks. This separation reduces overestimation while preserving the overall structure of DQN.

D. Lagrangian Formulation for Transmit Power Control

We formulate the mmWave transmit power-control problem as a constrained Markov decision process (CMDP). Let π_θ denote a parameterized policy with parameter vector θ . The expected discounted reward and cost are

$$J_R(\pi_\theta) = \mathbb{E}_{\tau \sim \pi_\theta} \left[\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t, s_{t+1}) \right], \quad (9)$$

$$J_C(\pi_\theta) = \mathbb{E}_{\tau \sim \pi_\theta} \left[\sum_{t=0}^{\infty} \gamma^t C(s_t, a_t, s_{t+1}) \right] \quad (10)$$

SNR constraint is captured via a cost of the form:

$$J_C(\pi_\theta) \leq d, \quad (11)$$

where d denotes the allowable cost threshold. Introducing a Lagrange multiplier $\lambda \geq 0$, the corresponding Lagrangian becomes

$$L(\theta, \lambda) = -J_R(\pi_\theta) + \lambda(J_C(\pi_\theta) - d). \quad (12)$$

The RL problem is solved via coupled primal-dual updates. The policy parameters follow the gradient step

$$\theta_{k+1} = \theta_k - \eta_k \nabla_{\theta} L(\theta_k, \lambda_k), \quad (13)$$

and the dual variable is updated as

$$\lambda_{k+1} = [\lambda_k + \zeta (J_C(\pi_{\theta_{k+1}}) - d)]_+, \quad (14)$$

where η_k is the primal learning rate, ζ is the dual stepsize, and $[\cdot]_+$ denotes projection onto the nonnegative orthant.

Following the analysis in [26], the curvature of $L(\theta, \lambda)$ with respect to θ is governed by $L(\lambda) = L_R + \lambda L_C$ where L_R and L_C are Lipschitz constants for the reward and cost gradients. Since $L(\lambda)$ grows as λ increases, the effective learning rate must shrink when the constraint becomes active. Two adaptive learning-rate rules are obtained by minimizing the primal error:

$$\eta_k^{\text{InvLin}} = \frac{1}{2L(\lambda_k)}, \quad (15)$$

$$\eta_k^{\text{InvQua}} = \frac{\mu}{2L(\lambda_k)^2}, \quad (16)$$

where μ denotes the strong-convexity parameter. Both rules automatically produce smaller learning rates when λ_k is large, improving stability under tight safety constraints, and larger learning rates when the constraint is inactive, improving convergence speed. In this work, we adopt this adaptive primal-dual structure to enforce SNR-based safety constraints during real-time mmWave transmit-power optimization, with λ_k governing constraint severity and η_k adapting accordingly.

IV. DEVELOPMENT SO FAR

In this section, we model the transmit power control problem as a constrained Markov decision process (CMDP) and solve it using deep reinforcement learning (DRL). The base station (BS) operates as the learning agent and interacts with the wireless environment through adjustments of its transmit power control parameters. Based on its observation of the current environmental state, the BS selects power control actions and receives corresponding rewards that reflect both power efficiency and communication reliability. The objective of the learning process is to maximize the expected cumulative reward while satisfying a predefined SNR constraint. **At this stage of development, the decision-making process is restricted to transmit power control, and beam selection is not yet incorporated into the learning framework.** The state space, action space, and reward function are defined as follows:

A. State

For transmit power control based on the base station's gain configuration, the base station (BS) adjusts its discrete gain parameters to regulate the transmit power P_{bs} . Since the received signal quality depends not only on the gain settings but also on the instantaneous channel conditions, the state is defined as the combination of the current gain configuration and the observed signal-to-noise ratio (SNR).

At time t , the state is represented as

$$s_t = [\mathbf{g}_t, \mathcal{S}_t], \quad (17)$$

where

$$\mathbf{g}_t = [g_{1,t}, g_{2,t}, \dots, g_{N,t}] \quad (18)$$

denotes the vector of quantized gain parameters, and γ_t represents the corresponding SNR measurement.

B. Action Space

The action space is discrete and consists of incremental adjustments to the gain parameters. At each time step, the agent modifies only one gain parameter g_i while keeping all others fixed. The possible actions correspond to increasing, decreasing, or retaining the current value of a single gain parameter. Let n be the total number of controllable gain parameters. The action space is then defined as

$$\mathcal{A} = \bigcup_{i=1}^n \{ \mathbf{e}_i^+, \mathbf{e}_i^-, \mathbf{0} \}, \quad (19)$$

where $\mathbf{e}_i^+$ and $\mathbf{e}_i^-$ represent a one-step increase or decrease in the i^{th} gain parameter, respectively, and $\mathbf{0}$ denotes no change.

C. Reward:

To evaluate the effect of the selected action a_t , both the channel's SNR and the transceiver gain parameters are considered. Let the base station (BS) gain configuration be

$$\mathbf{g}_t \triangleq [g_{1,t}, g_{2,t}, \dots, g_{N,t}],$$

where each gain parameter $g_{k,t}$ lies within its hardware-allowed range $[g_k^{\min}, g_k^{\max}]$. Since the gain parameters represent power-control values, they are converted to *linear scale* before normalization to ensure physically consistent weighting across components. The normalized linear-scale gains are defined as

$$\tilde{g}_{k,t} \triangleq \frac{g_{k,t}^{(\text{lin})} - g_k^{\min,(\text{lin})}}{g_k^{\max,(\text{lin})} - g_k^{\min,(\text{lin})} + \varepsilon} \in [0, 1], \quad (20)$$

where $g_{k,t}^{(\text{lin})}$ denotes the linear-scale value of the k^{th} gain parameter and $\varepsilon > 0$ prevents division by zero when needed.

We further define $\mathbf{w} \triangleq [w_1, w_2, \dots, w_N]$ as the nonnegative parameter weights satisfying $\sum_{k=1}^N w_k = 1$. To determine the relative contribution of each gain to the observed SNR, the absolute Pearson correlation coefficients are computed over the historical sequences $\{g_{k,\tau}\}_{\tau=1}^T$ and $\{\gamma_\tau\}_{\tau=1}^T$ as

$$c_k \triangleq |\text{corr}([g_{k,\tau}]_{\tau=1}^T, [\gamma_\tau]_{\tau=1}^T)| \geq 0. \quad (21)$$

The normalized weights are then obtained as

$$w_k \triangleq \begin{cases} \frac{c_k}{\sum_{j=1}^N c_j}, & \text{if } \sum_{j=1}^N c_j > 0, \\ \frac{1}{N}, & \text{otherwise (uniform weighting).} \end{cases} \quad (22)$$

Finally, the correlation-weighted power usage score penalizes large normalized gains quadratically:

$$J_{R_t} \triangleq \sum_{k=1}^N w_k \tilde{g}_{k,t}. \quad (23)$$

This quadratic penalty ensures that higher gain levels contribute disproportionately to the total power score, encouraging the agent to prefer minimal yet effective gain settings.

D. Cost

The safety constraint ensures that the received signal-to-noise ratio (SNR) remains above a predefined threshold required for reliable communication. Let $\mathcal{S}_t$ denote the observed SNR (in dB) at time step t . The SNR deficit relative to the threshold is defined as

$$\Delta_t \triangleq \max(0, \mathcal{S}_{\text{th}} - \mathcal{S}_t), \quad (24)$$

which is strictly positive only when the threshold is violated.

To obtain a smooth, differentiable measure of this violation, we employ a Gaussian-shaped cost function of the form

$$J_{C_t} = \exp\left(-\frac{\Delta_t^2}{2\sigma^2}\right), \quad (25)$$

where σ is a tunable sharpness parameter controlling how quickly the penalty grows as γ_t falls below the threshold. Here, the cost threshold is set to $d = 1$, ensuring that any deviation from the target is reflected as a normalized Gaussian penalty.

E. Lagrangian

Using the reward definition in (23) and the Gaussian SNR cost in (25), the corresponding Lagrangian for the constrained formulation is

$$L(\theta, \lambda) = -\mathbb{E}_{\pi_\theta} \left[\sum_{t=0}^{\infty} \gamma^t \sum_{k=1}^N w_k \tilde{g}_{k,t} \right] + \lambda \left(\mathbb{E}_{\pi_\theta} \left[\sum_{t=0}^{\infty} \gamma^t \exp\left(-\frac{\Delta_t^2}{2\sigma^2}\right) \right] - 1 \right) \quad (26)$$

where $d = 1$ is the cost threshold, and $J_R(\pi_\theta)$ and $J_C(\pi_\theta)$ are the discounted returns obtained from the per-step quantities J_{R_t} and J_{C_t} defined in (23) and (25), respectively.

The APD-DQN procedure in Algorithm 1 optimizes the gain-based power-control policy using the reward and cost structures defined earlier. At each time step, the agent observes the updated gain vector and SNR, forms the state, and selects an action either via ε -greedy exploration or by maximizing the Q-value. The updated gains are normalized to compute the instantaneous reward J_{R_t} in (23), while the SNR deficit Δ_t yields the Gaussian-shaped safety cost J_{C_t} in (25). These terms combine in the per-step Lagrangian reward $r_t = -J_{R_t} -$

Algorithm 1: APD-DQN for Gain-Based Power Control

Input: B : mini-batch size
 α : learning rate
 γ : discount factor
 $|\mathcal{D}|$: replay buffer capacity
 J : number of episodes
 C : target update period
 ε : exploration rate
 η_λ : dual learning rate
 d : cost threshold

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1 Initialize:  $\mathcal{D} \leftarrow \emptyset$ ; initialize  $Q_\theta, Q_{\theta^-}$ ;  $\lambda \leftarrow 0$ 
2 for  $episode = 1$  to  $J$  do
3   Observe  $\mathbf{g}_0, \gamma_0$ ;  $s_0 \leftarrow [\mathbf{g}_0, \gamma_0]$ 
4   for  $t = 0$  to  $T - 1$  do
5      $a_t \leftarrow \begin{cases} \text{random,} & \text{w.p. } \varepsilon \\ \arg \max_a Q_\theta(s_t, a), & \text{otherwise} \end{cases}$ 
6     Apply  $a_t$ ; observe  $\mathbf{g}_{t+1}, \gamma_{t+1}$ ;
        $s_{t+1} \leftarrow [\mathbf{g}_{t+1}, \gamma_{t+1}]$ 
7     Compute  $\tilde{\mathbf{g}}_{t+1}$  (normalized gains);
8     Compute  $J_{R_{t+1}} = \sum_{k=1}^N w_k \tilde{g}_{k,t+1}$  as in (23);
9     Compute  $\Delta_{t+1} = \max(0, \mathcal{S}_{th} - \mathcal{S}_{t+1})$ ;
10    Compute  $J_{C_{t+1}} = \exp\left(-\frac{\Delta_{t+1}^2}{2\sigma^2}\right)$  as in (25);
11     $r_t \leftarrow -J_{R_{t+1}} + \lambda(J_{C_{t+1}} - d)$ 
12    Store  $(s_t, a_t, r_t, s_{t+1}, J_{C_{t+1}})$  in  $\mathcal{D}$ 
13    if  $|\mathcal{D}| \geq B$  then
14      Sample  $\{(s_i, a_i, r_i, s'_i, c_i)\}_{i=1}^B \sim \mathcal{D}$ 
15      for  $i = 1$  to  $B$  do
16         $y_i \leftarrow r_i + \gamma Q_{\theta^-}(s'_i, \arg \max_a Q_\theta(s'_i, a))$ 
17         $\theta \leftarrow \theta - \alpha \nabla_\theta \frac{1}{B} \sum_{i=1}^B (Q_\theta(s_i, a_i) - y_i)^2$ 
18         $\bar{c} \leftarrow \frac{1}{B} \sum_{i=1}^B c_i$ 
19         $\lambda \leftarrow [\lambda + \eta_\lambda(\bar{c} - d)]_+$ 
20    if  $t \bmod C = 0$  then
21       $\theta^- \leftarrow \theta$ 

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$\lambda(J_{C_t} - d)$, consistent with the overall Lagrangian form in (26). Transitions are stored in the replay buffer, and mini-batches are used to update the Q-network via standard DQN bootstrapping. The dual variable is updated using the empirical batch-average cost, $\lambda \leftarrow [\lambda + \eta_\lambda(\bar{c} - d)]_+$, enforcing the safety constraint defined by the threshold d . Periodic updates of the target network stabilize learning, allowing the coupled primal–dual dynamics to adjust transmit gains while maintaining SNR safety.

V. NEXT STEPS OF DEVELOPMENT

The next stage of this work aims to extend the current transmit power control framework by incorporating transmit and receive beam selection into the learning process. Joint optimization of transmit power and beam selection is expected

to further improve link efficiency by exploiting directional gains available in mmWave systems under mobility.

However, introducing beam selection substantially increases the dimensionality of the decision space, as the learning agent must select from a significantly larger set of discrete beamforming options in addition to the existing power control parameters. This expansion of the action space leads to a combinatorial growth in the number of possible action configurations, which in turn increases the complexity of the learning problem.

As a consequence, naively extending the current reinforcement learning framework to include beam selection would result in a significant increase in training overhead, including longer convergence times and higher sample complexity. Addressing these challenges—through structured action representations, efficient exploration strategies, or reduced-complexity beam search mechanisms—constitutes the focus of the next stage of this work.

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